

The machine-readable parameter block (the YAML front matter of docs/display_transform.md, read by src/comfortzone/display.py):

```
# Machine-readable part. `src/comfortzone/display.py` reads THIS block (the YAML between the first
# two '---' lines); change a value here and every script that uses the transform follows.
transform: display_minification
version: 1
date: 2026-09-25
applies_to:
  - study1_cutin_video # stimuli shown as video on a participant's own desktop monitor
  - study2_cutin_video # first crowd-sourced study (Random, Sequence and Button paradigms)
  - study2_cutin_video # second crowd-sourced study
does_not_apply_to:
  - highD # optical input was the real world
  - inD
  - test_track
parameters:
  screen_diagonal_in: 23.0 # monitor diagonal [inch]; "typically a standard 23-inch screen"
  screen_aspect_w: 16 # monitor aspect ratio, width part
  screen_aspect_h: 9 # monitor aspect ratio, height part
  viewing_distance_cm: 60.0 # eye to screen [cm]
  virtual_hfov_deg: 90.0 # horizontal field of view of the rendering camera [deg]
  image_width_fraction: 1.0 # shown image width / screen width (1.0 = full-screen video)
  camera_to_front_m: 0.0 # rendering camera to the ego's front bumper [m]; 0.0 keeps the
  # pipeline's convention (card EL.1b: the observer at the bumper)
switch:
  env_var: CZB_DISPLAY_TRANSFORM
  values: ["off", "on"]
  default: "off" # "off" reproduces every result so far exactly
```

The display transform: from rendered kinematics to what the participants saw

Written 2026-09-25 (night) at Jonas's request. The YAML block above is the single source of the parameters; the code is src/comfortzone/display.py, its property checks tests/test_display.py, and the comparison with the untransformed analyses is card DT.1 (replication/czb/dt1_display_transform.py → replication/czb/out/dt1_display_transform.md).

1 The problem

The participants of the two crowd-sourced cut-in studies watched rendered clips on a desktop monitor (about 60 cm away, typically a 23-inch 16:9 screen). The rendering camera had a 90° horizontal field of view; the monitor, at that distance, fills only about 46° of the participant's. The picture is therefore **minified**: every object subtends a smaller visual angle, and expands more slowly, than it would have from the driver's seat of the virtual world. Every optical variable we computed so far (the angular size θ and the looming rate $\dot{\theta}$ of card EL.1b) is the *rendered world's*, not the *participant's*. Real-world data (highD, inD, the test track) carry the real optical variables. To compare the two, the video's variables must be put in the participant's terms.

2 The geometry (exact)

Notation, all derived from the parameters above:

symbol	definition	value with the defaults
S_w	screen width = diagonal $\times 2.54 \times a_w / \sqrt{a_w^2 + a_h^2}$ [cm]	50.92 cm
I_w	image width = image_width_fraction $\times S_w$ [cm]	50.92 cm
F	the rendering's focal length in screen units = $(I_w / 2) / \tan(\text{HFOV}_v / 2)$ [cm]	25.46 cm
d	viewing distance [cm]	60 cm
k	the display gain F / d	0.4243
HFOV_d	the horizontal angle the image fills for the viewer = $2 \operatorname{atan}(I_w / (2d))$	45.98°

A pinhole camera maps a scene point at camera coordinates (X forward, Y lateral, Z up) to the screen point ($F \cdot Y/X$, $F \cdot Z/X$). A viewer whose eye sits at distance d on the perpendicular through the image centre sees that screen point in the direction

$$\tan \beta_Y = (F/d) \cdot Y/X = Y / (X/k), \quad \tan \beta_Z = Z / (X/k).$$

This is **exactly** the direction in which the viewer would see a scene point at (X/k, Y, Z). So:

The participant's optical array is that of an equivalent world in which every distance along the line of sight is multiplied by 1/k and nothing else changes (lateral and vertical coordinates, object sizes, lateral speeds unchanged).

It holds at every azimuth, not only near the centre, and needs no small-angle approximation. Assumptions: a flat screen; the eye centred in front of the image at distance d; square pixels (the vertical field of view follows from the horizontal one); monocular optics (binocular disparity, accommodation and the screen's own flatness cues are not modelled; they all say "a flat picture 60 cm away" and are ignored here, as in the display literature's geometric-field-of-view account).

3 The transform of the kinematic variables

With $r = \text{gap} + \text{camera_to_front_m}$, the camera-to-target distance along the line of sight, and dv the closing speed ($dr/dt = -dv$):

variable	rendered world (so far)	as perceived (transform "on")	ratio perceived/rendered
distance r	r	$r_p = r / k$	$1/k = 2.36$
closing speed dv	dv	$dv_p = dv / k$	$1/k = 2.36$
lateral position, lateral clearance l_0 , lateral rate	$y, l_0, \dot{l}$	unchanged	1
target width W	W	unchanged	1
angular size $\theta = 2 \operatorname{atan}(W / 2r)$	θ	$\theta_p = 2 \operatorname{atan}(W k / 2r)$	$\approx k$ (exactly k in the small-angle limit)
looming $\dot{\theta} = W \cdot dv / (r^2 + W^2/4)$	$\dot{\theta}$	$\dot{\theta}_p = W \cdot dv_p / (r_p^2 + W^2/4) = k \cdot W \cdot dv / (r^2 + k^2 W^2/4)$	$\approx k$; exactly $k \cdot (r^2 + W^2/4) / (r^2 + k^2 W^2/4)$
TTC = r / dv	TTC	$TTC_p = r_p / dv_p = TTC$	1 (invariant)
inverse tau $\theta/\dot{\theta}$	τ^{-1}	$\approx \tau^{-1}$ (exact in the small-angle limit)	≈ 1
accelerations along the line of sight	a	a / k	2.36

With $\text{camera_to_front_m} = 0$ (the default) the perceived variables are obtained by feeding the pipeline's own formula (card EL.1b) with **gap / k and dv / k**. Off-axis (a target at lateral offset y), the rear face's perceived angular width is $\operatorname{atan}((y + W/2)/r_p) - \operatorname{atan}((y - W/2)/r_p)$; the pipeline has always used the centred form, and this transform does not change that choice (`display.py` offers the off-axis form for later use).

4 What this does and does not do to the results — read this before using it

- Looming-based levels shrink by $k \approx 0.42$.** A comfort-zone level fitted on the video's looming, e.g. the intervention level 0.0320 rad/s (card JJ.10), becomes about 0.0136 rad/s of *perceived* looming. This is the level to compare with real-world looming (highD's 0.0388; Farewell's 0.02).
- TTC and inverse tau do not change.** Optical tau is invariant under magnification; the equivalent world scales distance and closing speed by the same factor. Every TTC-based result (the hard boundary on TTC, card NC.3j's 3.3 s, card NC.3l's 1.64 s, the TTC comparators) stands as it is. The transform does NOT, by itself, "offset things toward higher TTCs".
- The offset toward higher TTC appears when the perceived looming is carried to a real driver at the TRUE closing speed.** A real driver closing at the video's dv on a car of width W receives the participant's perceived looming $\dot{\theta}_p$ at the distance $r_b = \sqrt{(W \cdot dv / \dot{\theta}_p - W^2/4)}$, i.e. at $TTC_b = r_b / dv \approx TTC / \sqrt{k} = \mathbf{1.54 \times TTC}$. That is convention (b) below, and it is the sense in which the video's boundary lies at a *higher real-world TTC* once the display is accounted for.
- Within-video model comparisons barely change.** Every probit-on-log-looming model absorbs a constant factor into its level ($\log \dot{\theta}_p = \log \dot{\theta} + \log k + \text{a small correction at short range}$), so held-out scores move only through that correction; spreads, gates (lateral, unchanged) and rank correlations (per-driver transfer, card TR.1) are unaffected.

Two conventions for "the equivalent real-world situation"

convention	what is matched	distance	closing speed	TTC	use it for
(a) optical (the default "on")	θ and $\dot{\theta}$ both, same W	r / k	dv / k	TTC	comparing optical variables (looming, angular size) with real-world data
(b) at true speed	$\dot{\theta}$ only, same W , true dv	$\sqrt{(W \cdot dv / \dot{\theta}_p - W^2/4)}$	dv	$\approx$ TTC $/ \sqrt{k}$	stating a video boundary as a real-world TTC at the video's own speeds

Convention (a) is exact about what reached the eye. Convention (b) assumes the participants knew the true closing speed from somewhere other than the optics (e.g. the ego's speed from the instructions); whether they did is not known, so (b) is reported beside (a), never in its place.

5 How to use it

- **Switch.** `CZB_DISPLAY_TRANSFORM=on` (environment) turns it on for code that asks `display.active_geometry()`; unset or off gives None and every function is the identity. Card DT.1 does not rely on the environment: it runs each analysis twice, explicitly.
- **Parameters.** Edit the YAML block at the top of this file, or pass overrides:
`display.load_geometry(viewing_distance_cm=70)`.
- **Functions** (`src/comfortzone/display.py`): `gain(g)`, `display_hfov_deg(g)`, `perceived(gap, dv, W, g) → dict of r_p, dv_p, θ_p, θ_p, TTC_p`; `looming(gap, dv, W, g=None)` (the EL.1b formula, perceived when g is given); `offaxis_theta(gap, y, W, g)`; `ttc_at_true_speed(theta_dot_p, dv, W)` (convention b); `transform_level(level, gap, dv, W, g)`.
- **Pre-registered scripts are not modified** (standing rule). New analyses that must hold for both take a geometry argument; DT.1 re-runs the existing ones by importing their functions.

6 Results (card DT.1, 2026-09-26)

`replication/czb/out/dt1_display_transform.md` and, over the parameters, `replication/czb/out/dt1b_display_sensitivity.md`; summary in `replication/czb/out/worklog.md` (2026-09-26). Within the video studies only the looming levels move ($\times 0.424$); against real data the comparisons with Farewell's emergency level, highD's level and the matching deceleration reverse.

Card DT.2 (the left turn, 2026-09-26) qualifies this. The test track (real optics) and the video agree (PET_50 2.45 against 2.42 s), and on video the left-turn answers follow distance or looming, not time. Under the transform a shared distance or looming criterion would put the video boundary outside the design (-1.4 to 0.4 s); the effective gain that reconciles track and video is 0.99 [0.84, 1.05], not 0.42. The participants judged as if the display had not shrunk the scene: the transform, as applied to their criterion, may over-correct. Report cut-in cross-domain numbers both ways until the per-participant display test decides (query DT2.Q1). Plain-language account with figures and a video: `docs/display_transform_explained.md`.

7 Open points for Jonas

- The parameters are the stated defaults; if the studies' logs carry screen resolution or window size, `image_width_fraction` is the one most likely to be below 1 (a video in a browser window).
- Where was the rendering camera (the driver's eye, the bumper)? `camera_to_front_m = 0` keeps the pipeline's convention; a driver's eye is about 1.5–2.5 m behind the bumper, which matters at the shortest gaps (TTC and looming both).
- Literature (to verify before citing): minifying the geometric field of view relative to the display field of view makes distances look larger and speeds faster in driving simulators (e.g. Diels & Parkes, 2010, on geometric field of view and perceived speed).